United International University

CSE 4510

Assignment-1

A computer system is running several processes. The processes arrive at different times and require different amounts of CPU time. The operating system uses a modified Round Robin scheduling algorithm to provide fair CPU sharing while giving preference to processes with shorter CPU burst times.

Consider the following set of processes:

Process	Arrival Time	Burst Time
P1	**0**	**7**
P2	**1**	**4**
P3	**3**	**6**
P4	**3**	**2**
P5	**5**	**3**
P6	**5**	**8**
P7	**8**	**5**
P8	**8**	**1**

Scheduling Rules:

Use Round Robin scheduling with a Time Quantum of 3 ms.

Also, this system follows the additional rule:

1. The Time Quantum is 3 ms.
2. At any scheduling point, consider all processes that have already arrived and are waiting in the ready queue.
3. Among the available processes, the process with the lowest remaining Burst Time gets selected first.

4. Therefore, if two or more processes become available at the **same time**, the process with the lower Burst Time must be selected first.
5. If a new process arrives while another process is executing, the currently running process will continue until:
    - its burst is completed, or
    - its 3 ms Time Quantum expires.
6. When the CPU needs to select another process, the process with the lowest remaining Burst Time among all ready processes gets priority.
7. If two processes have the same remaining Burst Time, the process with the earlier Arrival Time is selected first. If both are also equal, select the process with the lower process number.
8. A process that does not finish within its Time Quantum goes back to the ready queue with its remaining burst time.
9. Newly arrived processes must be considered when making the next scheduling decision.

Tasks:

a. Draw the complete Gantt Chart.

b. Calculate the Completion Time (CT) of every process.

c. Calculate the Turnaround Time (TAT) of every process.

d. Calculate the Waiting Time (WT) of every process.

e. Calculate the Average Waiting Time.

f. Calculate the Response Time (RT) of every process.

g. Calculate the Average Response Time.

h. Calculate the Total Execution Time required to complete all processes.

Write a program to implement the above modified Round Robin scheduling algorithm.